Problem1

a. independent.
Numerically: P(B,E)=P(B)P(E)
Topologically : B and E don't have edges. one doesn't have direct influence on the other.
b.
According to the Bayes' Rule,

P(B,E| a)= $\frac{P(a|B,E)P(B,E)}{P(a)}$,

From table we could find the

B	E	P(B,E| a)
T	T	95*0.001*0.002=0.00019
t	F	94*0.001*0.998=0.093812
f	T	29*0.999*0.002=0.057942
f	f	0.1*0.999*0.998=0.0997

Normalization
0.251644

B	E	P(B,E| a)
T	T	0.0008
t	F	0.372796
f	T	0.23025
f	f	0.39619

P(B|a) = P(B,E=true|a) +P(B,E=false |a) =0.0008+0.3728=0.3736
P(E|a) = P(B = true,E|a) +P(B=false,E |a) 0.0008+0.2303=0.2311
P(B,E| a) = 0.0008 != P(E|a) * P(B|a) = 0.3736*0.2311=0.086339

Problem 2

a.

network **c claim that** P(Gfather , Gmother , Gchild) = P(Gfather)P(Gmother)P(Gchild).
Because three node is separated. Three variable are independent.
We have equation :
P(Gfather , Gmother , Gchild) = P(Gfather | Gmother, Gchild) P(Gmother |Gchild) P(Gchild).
Thus, when three variable are independent, that is to say, there is no links among them.
We have :
P(Gfather) =P(Gfather | Gmother, Gchild),
P(Gmother |Gchild) = P(Gmother)

So we can find:
P(Gfather , Gmother , Gchild) = P(Gfather)P(Gmother)P(Gchild).

b.

(A) and (b) networks make independence claims that are consistent with the hypothesis about the inheritance of handedness. (C) ignores that the gene itself is equally likely to be inherited from either of an individual's parents.

c

(a) is best. (b) has extra links , Hfather and Hmother link to Hchild. These two directed edges are not described in the hypothesis.

D

CPT condition possibilities table
突变概率 为 m

Gmother	Gfather	P(Gchild =l)	P(Gchild = r)
L	R	0.5	0.5
L	L	1-m	M
R	R	m	1-m
R	L	0.5	0.5

E

Gmother	Gfather	Possibilities	P(Gchild =l)	P(Gchild = r)
L	R	Q*(1-q)	0.5* Q*(1-q)	0.5
L	L	Q*q	(1-m)* Q*q	M* Q*q
R	R	(1-q)*(1-q)	M*(1-q)*(1-q)	(1-m) *(1-q)*(1-q)
R	L	q*(1-q)	0.5* Q*(1-q)	0.5

(1-m)* Q*q + Q*(1-q) + M*(1-q)*(1-q) = q^2-mq^2 +q –q^2 +M(1-2q+q*^2)
= q+m-2mq

F

Genetic equilibrium means that P(Gchild) = P(Gparent) = P(Gmother) = q
Q = q + m – 2mq
q = 0.5

However, we already know that most people are accustomed to using the right hand instead of half using the right hand, so the hypothesis at the beginning of this question must be wrong.

Problem3

DBN
P(X0) = 0.7
前一晚睡 enough，这一晚睡 enough 概率为 0.8
前一晚没睡 enough，这一晚睡 enough 概率为 0.3
睡 enough,学生红眼概率为 0.2
睡 enough,学生红眼概率为 0.2

条件		红眼	Sleep in class	后一晚 sleep enough
Enough sleep	True0.7	0.2	0.1	0.8
Enough sleep	False0.3	0.7	0.3	0.3

HMM
只有一个随机变量. 我们可以把 DBN 的所有状态变量合成成一个单一的状态变量，取值为单个状态变量所有可能的取值组成的元组.

条件		红眼,上课睡觉	红眼,上课不睡觉	不红眼,上课不睡觉	不红眼,上课睡觉
Enough sleep	True0.7	0.2*0.1=0.02	0.2*0.9=0.18	0.8*0.9=0.72	0.8*0.1=0.08
Enough sleep	False0.3	0.7*0.3=0.21	0.7*0.7=0.49	0.3*0.7=0.21	0.3*0.3=0.09

Problem4

a.

$$\mathrm{P(EnoughSleep_0) = P(X_0) = 0.7}$$
$$\mathrm{P(EnoughSleep_1)} = P(X_1) = 0.8 * 0.7 + 0.3 * 0.3 = 0.65$$

$$\mathrm{P(X_1|e_1)} = \mathrm{P(EnoughSleep_t|e_{1:}}_t) = \alpha \mathrm{P(e_1|X_1)P(X_1|e_0)} = \alpha \mathrm{P(e_1|X_1)P(X_1)} = \alpha$$
$$< 0.72 * 0.65 \ ,0.21 * 0.35 \ > = < 0.8643,0.1357 >$$
$$\mathrm{P(EnoughSleep_2|e_{1:2})} = \alpha \mathrm{P(e_2|X_2)P(X_2|e_{1:1})} = \alpha \mathrm{P(e_2|X_2)} \sum_{x_t} \mathrm{P(X_2|X_1}, e_{1:t})\mathrm{P(X_1|}e_{1:t})$$

$$利用马尔可夫性质可得 = \alpha P(e_2|X_2)\sum_{x_t} P(X_2|x_1)P(X_t|e_{1:t})因为X_2和\ e_{1:t}无关.P(X_2|e_1)$$

$$= \sum_{x_1} P(X_2|x_1)P(X_1|e_1) = 0.8643 * 0.8 + 0.1357 * 0.3$$

$$= 0.73215 \qquad 第一天睡好了,第二天睡好\ 0.8\ ,第一天没睡好,第二天睡好\ 0.3$$

$$P(EnoughSleep_2|e_{1:2}) = \alpha P(e_2|X_2)P(X_2|e_1) = \ 0.73215 * 0.18 = 0.131787$$

E2 为红眼，没有睡觉，$P(e_2|X_2) = \ < 0.18,0.49 >$

$$P(EnoughSleep_2|e_{1:2}) = \alpha P(e_2|X_2)P(X_2|e_1) = 0.2679 * 0.49 = 0.131271$$

归一化

0.131271/(0.131271+0.131787)=0.4990

综上所述，

$$P(EnoughSleep_2|e_{1:2}) = \ < 0.5010,0.4990 >$$

E3 为红眼，睡觉

$$P(e_3|X_3) = \ < 0.02,0.21 > 根据\ problem3\ 的表可以得到$$

$$P(EnoughSleep_3|e_{1:3}) = \alpha P(e_3|X_3)P(X_3|e_{1:2}) \quad 这个公式记忆就记住凑起来是条件e_{1:3}$$

$$= \ \alpha P(e_3|X_3)\sum_{x_t} P(X_3|X_2, e_{1:2})P(X_2|e_{1:2}) \quad 向前传播$$

$$P(X_3|e_{1:2}) = \sum_{x_2} P(X_3|x_2)P(X_2|e_{1:2}) = 0.5010 * 0.8 + 0.4990 * 0.3 = 0.5505$$

$$第一天睡好了,第二天睡好\ 0.8\ ,第一天没睡好,第二天睡好\ 0.3$$

$$P(EnoughSleep_3|e_{1:3}) = \alpha P(e_3|X_3)P(X_3|e_{1:2}) = \ < 0.02 * 0.5505 = 0.01101,0.21 * 0.4495 = 0.094395 >$$

$$归一化后\ \ < \frac{0.01101}{0.01101 + 0.094395} = 0.1045,0.8955 >$$

b

$$P(EnoughSleep_t|e_{1:3})$$

Smoothing 平滑是用已知证据计算过去的状态分布.

$$P(X_k|e_{1:t}) = \alpha P(X_k|e_{1:k})P(e_{k+1:t}|X_k)$$

$$P(e_3|X_3) = \ < 0.02,0.21 > 根据\ problem3\ 的表可以得到$$

$P(e_3|X_2) = \sum_{x_3} P(e_3|x_3)P(\ \ |X_3)P(X_3|X_2) =< 0.02 * 0.8 + 0.21 * 0.2,0.02 * 0.3 + 0.21 * 0.7 > \ =< 0.0588,0.153 >$

解释上面的式子:

睡 enough，第二天 0.8 睡 enough 然后观察到的概率为 0.02，第二天 0.2 睡不 enough 然后观察到的概率为 0.21，

睡不 enough，第二天 0.3 睡 enough，观察到概率为 0.02…

E2 为红眼，没有睡觉，$\mathrm{P}(\mathrm{e}_2|\mathrm{X}_2) = < 0.18,0.49 >$

$$\mathrm{P}(\mathrm{e}_{2:3}|\mathrm{X}_1) = \sum_{x_2} \mathrm{P}(\mathrm{e}_2|\mathrm{x}_2)\mathrm{P}(e_3|\mathrm{X}_2)\mathrm{P}(\mathrm{X}_2|X_1) =< 0.8 * 0.0588 * 0.18 + 0.2 * 0.153 * 0.49$$
$$= 0.023461,0.3 * 0.0588 * 0.18 + 0.7 * 0.153 * 0.49 = 0.055654 >$$

解释上面的式子:
第一项和第三项根据模型直接得到，第二项是之前计算出来的.
睡 enough，第二天 0.8 睡 enough 然后观察到的概率为, 0.0588*0.18
第二天 0.2 睡不 enough 然后观察到的概率为 0.153*0.49
睡不 enough，第二天 0.3 睡 enough，观察到概率为 0.588*0.18
睡不 enough，第二天 0.7 睡不 enough，观察到概率为 0.153*0.49
上面是后向消息

然后我们之前在第一问 a 计算了 前向滤波

$$\mathrm{P}(X_k|e_{1:t}) = \alpha P(X_k|e_{1:k})P(e_{k+1:t}|X_k)$$
$$\mathrm{P}(X_1|e_{1:3}) = \alpha P(X_1|e_1)P(e_{2:3}|X_1) =< 0.8643 * 0.023461 = 0.020277,0.1357 * 0.055654$$
$$= 0.007552 > \textit{归一化后} < \frac{0.020277}{0.020277 + 0.007552} = 0.728628,0.2713 >$$

$$\mathrm{P}(\mathrm{EnoughSleep}_2|\mathrm{e}_{1:2}) = < 0.5010,0.4990 >$$

$$\mathrm{P}(X_2|e_{1:3}) = \alpha P(X_2|e_{1:2})P(e_3|X_2) =< 0.5010 * 0.0588 = 0.029459,0.4990 * 0.153$$
$$= 0.076347 > \quad \textit{归一化后} < \frac{0.029459}{0.029459 + 0.076347} = 0.2784,0.7216 >$$

$$\mathrm{P}(X_3|e_{1:3}) = \alpha P(X_3|e_{1:3})P(\quad|X_3) = \textit{上面计算出来的}\,\mathrm{P}(\mathrm{EnoughSleep}_3|\mathrm{e}_{1:3})$$
$$= \alpha\mathrm{P}(\mathrm{e}_3|\mathrm{X}_3)\mathrm{P}(\mathrm{X}_3|\mathrm{e}_{1:2}) = \text{归一化后} < \frac{0.01101}{0.01101 + 0.094395}$$
$$= 0.1045,0.8955 >$$

C

t = 1，filter $< 0.8643,0.1357 >$
smooth$< 0.728628,0.2713 >$
t =2

$$\mathrm{P}(\mathrm{EnoughSleep}_2|\mathrm{e}_{1:2}) = < 0.5010,0.4990 >$$

smooth
$<0.2784,0.7216 >$
smoothing 更相信学生缺乏睡眠